

Evolution of man

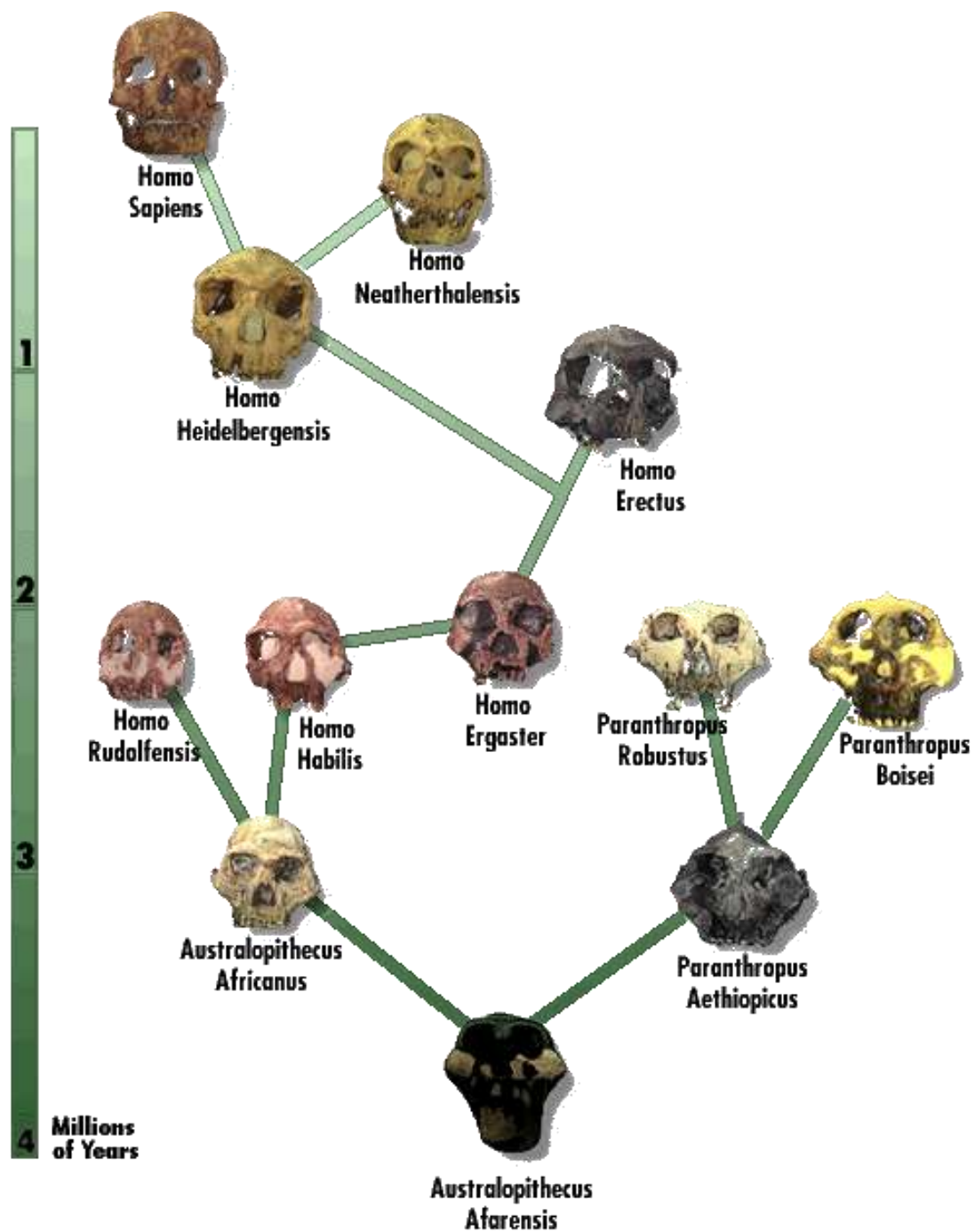
B.Sc. 1st Semester – Paper 1A (Unit 5)

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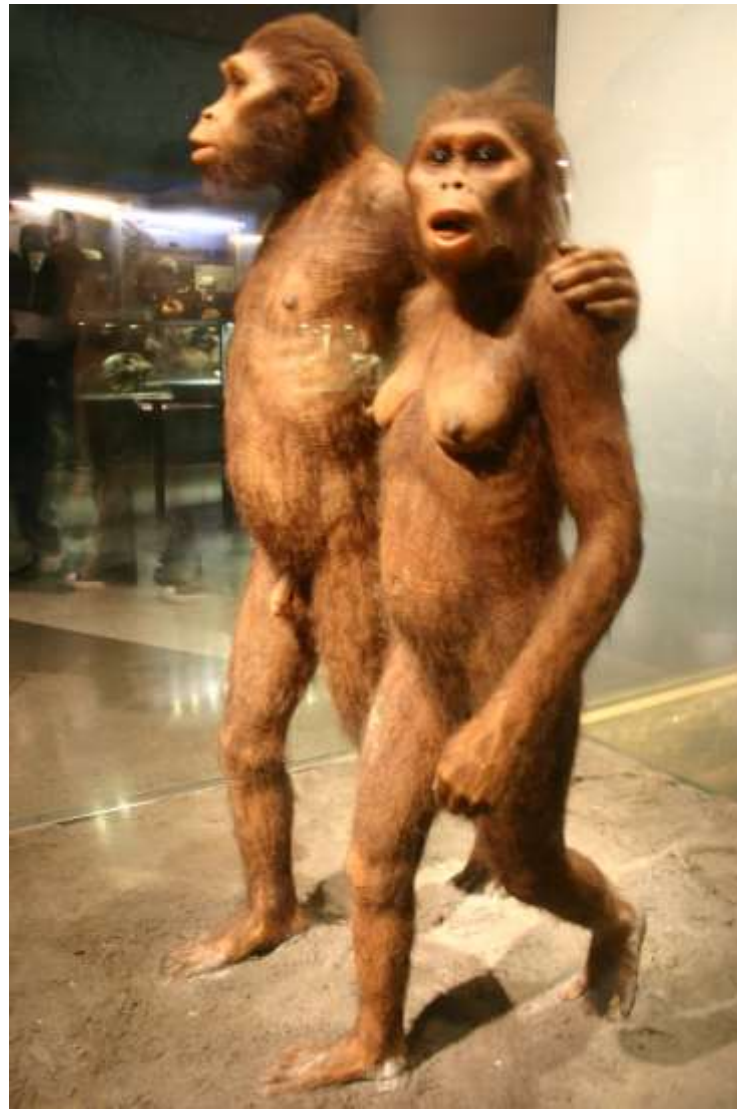
- Human evolution (Mode of evolution: Phyletic gradualism where a species A gradually transforms into a species B)
- It has been standard practice to put Homo sapiens at the end of a continuous series tracing back through at least two other species.

- Ramapithecus - These are ancestral forms (apes) living approx. 8 – 14 Mya.
- They had a flattened face with a short muzzle.
- First fossil (a fragile upper jaw) was recovered by G. Edward Lewis in 1930 from Northern India. Another fossil was recovered from Lake Victoria in Africa by Louis Leaky 1955.
- Fossil evidence indicate that these were the forerunners of Hominids.

- Australopithecus – The famous fossil of a female ape-man called Lucy was recovered from Ethiopia by Donald Johanson in 1973.
- This is probably the oldest fossil among all primitive man described.
- This group lived 1.5 – 5 Mya. They had a flattened face, an upright gait, a heavy skull and dentition.
- Two reported groups: A. robustus and A. africanus

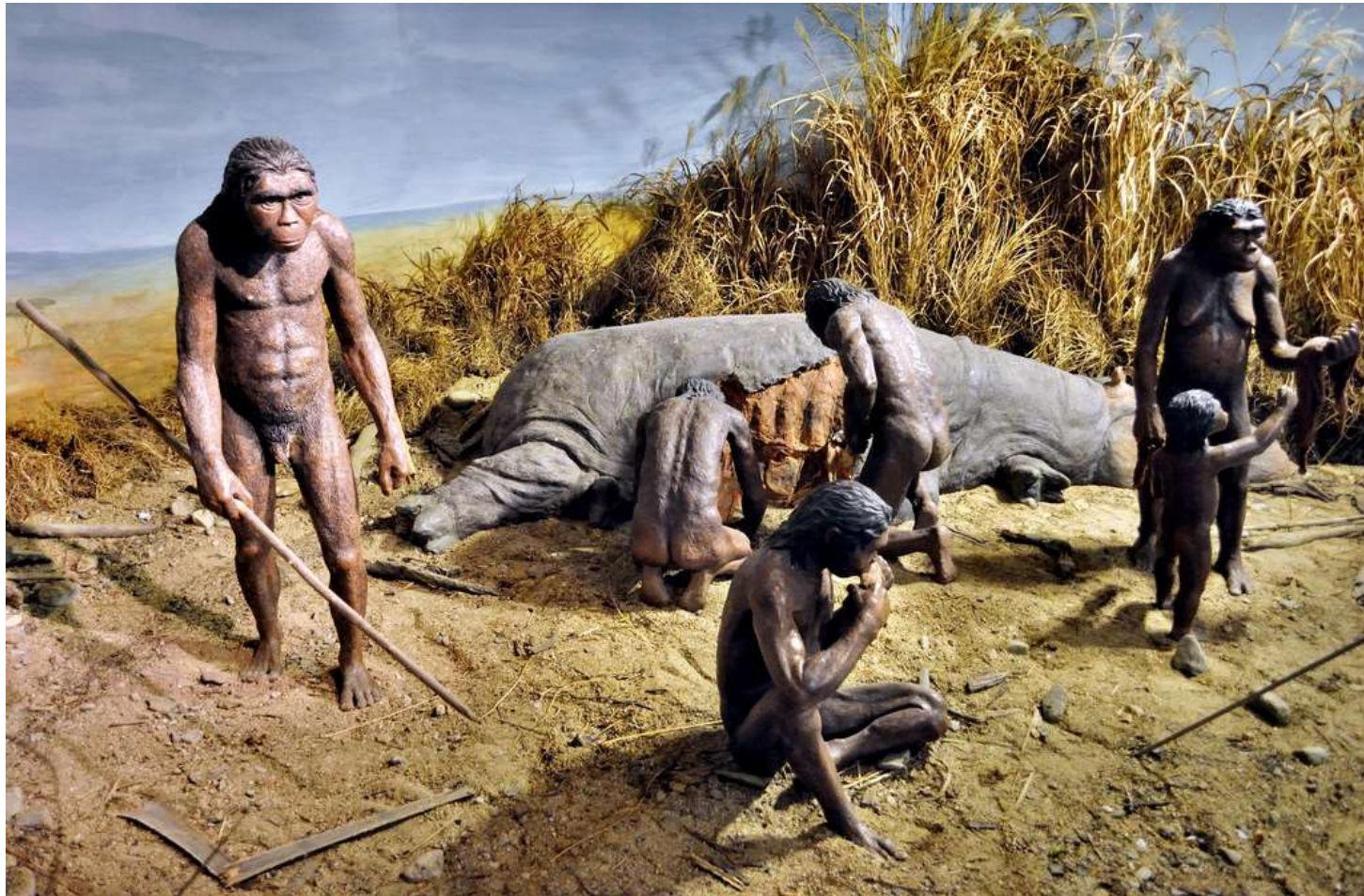
Australopithecus

(L. *austral* = southern + Gr. *Pithekos* = ape)



- Homo habilis – Also known as the “handy man”
- Fossil remains recovered by Leaky in 1964 in East Africa.
- 1.5 – 2 Mya.
- Erect gait, relatively large brains, tool users, fabricated crude tools out of pebbles and used to gather a variety of plants and small animals.

Homo habilis



- Homo erectus – upright walking man, Java men, Peking men (peking fossils discovered by Davidson Black, 1920s).
- 1 Mya
- Fossil remains were discovered at Trinil, Java, in 1894 by Eugene Dubois.
- They used tools of bone and stone and used fire.
- Had a cranial capacity of 775 – 1000 cc (Peking men = 900 – 1200 cc)

Homo erectus



- Homo sapiens neanderthalensis – Fossils were first discovered in 1856 in a limestone cave in Germany.
- 1.5 lakh years years old cave dwellers.
- Short (5 feet), powerfully built, prominent facial brow ridges, large brains (1450 cc)
- Used complex stone tools, were accomplished hunters, used animal hides as clothing.

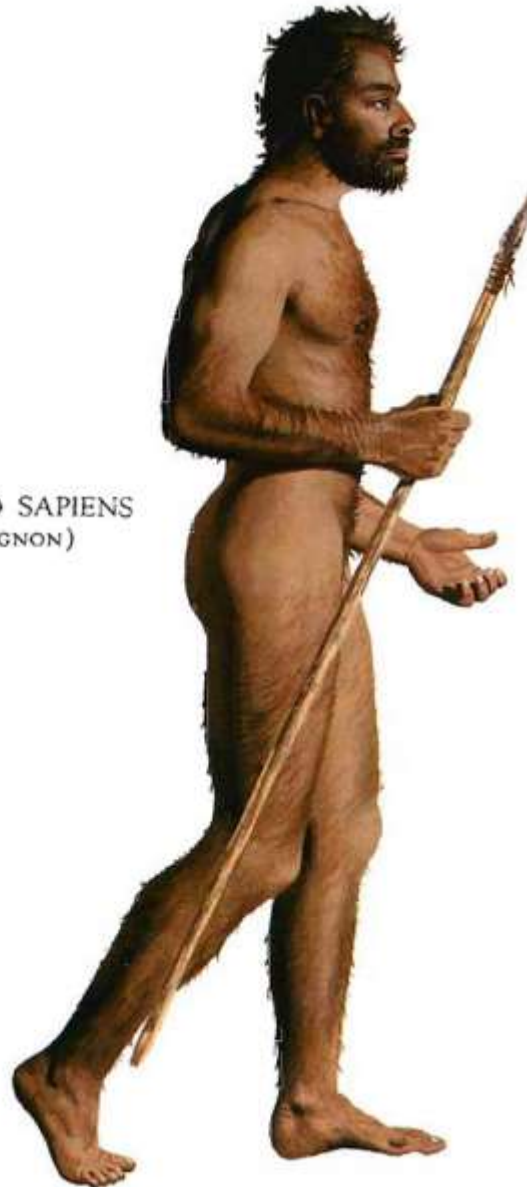
Neanderthals



- Homo sapiens sapiens – Cro-magnon man
- Earliest Fossil remains were unearthed from a French rock shelter.
- 35,000 years old.
- They occurred in Europe and left behind elaborate cave paintings.

Cro-magnon man

HOMO SAPIENS
(CRO-MAGNON)



- Homo sapiens – Modern man
- 1350 cc of cranial capacity.
- Origin – Asia or Africa?

- Homo naledi - ***Homo naledi*** is an extinct species of hominin, which anthropologists first described in 2015 and have assigned to the genus Homo.
- In 2013, fossil skeletons were found in South Africa's Gauteng province, in the Rising Star Cave system, part of the Cradle of Humankind World Heritage Site.
- As of 10 September 2015, fossils of at least fifteen individuals, amounting to over 1550 specimens, have been excavated from the cave.



Conclusion

